

A Secure and Scalable Framework for Student Recruitment Using Blockchain and AI

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Abstract: Campus recruitment is often bogged down by repetitive data entry, inefficient workflows, and the constant challenge of verifying student documents. To address these bottlenecks, we are developing the Student Placement and Coordination Platform (SPCP). Our goal is to create a unified system that blends the security of blockchain with the efficiency of Artificial Intelligence. While the blockchain layer for immutable verification is currently in the design phase, we have successfully implemented and tested the platform's AI core. Using a combination of LangChain and LangGraph powered by the Gemini API, we built an intelligent agent capable of understanding and organizing complex resume data. This paper details our architectural vision and presents experimental results from our AI module, demonstrating how we can transform unstructured student data into actionable insights to streamline the hiring process.

Keywords: Blockchain, SaaS (Software as a Service), Campus Placement, Credential Verification, Student Recruitment, Campus Automation, Machine Learning, Decentralized Verification.

1 INTRODUCTION

1.1 Background and Motivation

The transition from college to the professional world via campus placement is critical for students and a key talent source for companies. Current systems, however, are manual and outdated, failing to meet the scale of modern hiring. These traditional methods cause data duplication, documentation delays, and opacity, frustrating both students and recruiters. Furthermore, the prevalence of fraudulent academic credentials adds a significant verification burden, making hiring more difficult.

1.2 Problem Statement

The objective is to create an effective, trustworthy, and reliable placement system. Students currently face redundant data entry across multiple portals, while recruiters are burdened with administrative work like manual screening, eligibility checks, and verification. Concurrently, placement officers lack the integrated, real-time analytics needed for strategic decision-making. Addressing these

conflicting needs requires an integrated, secure, and smart platform to modernize campus recruitment.

1.3 Proposed Solution and Contributions

This study introduces the Student Placement and Coordination Platform (SPCP), which integrates decentralized and intelligent technologies. The platform's innovations include:

Blockchain Credentials Service: Provides immutable, instantly verifiable student records to prevent fraud and verification delays.

AI-powered Matching: Uses ML pipelines to parse resumes and provide personalized job suggestions, making shortlisting more effective.

Unified, Multi-tenant Architecture: Offers students a single digital profile while providing separate, analytical dashboards for recruiters and administrators.

2 LITERATURE REVIEW

The integration of blockchain technology into the validation of academic credentials has recently

garnered considerable attention, with numerous studies examining its potential to enhance transparency, authenticity, and security in educational certification processes.

Noshi and Xu (2024) proposed a Blockchain-Based Academic Credential Verification System to combat traditional verification inefficiencies and fraud. Their prototype uses blockchain's immutable ledger for the secure storage and verification of academic records. This approach expedites the process, reduces manual involvement, and builds trust between employers and issuing organizations. [1]

Rustemi et al. (2023) conducted a comprehensive literature review on blockchain systems for academic certificate verification. Their study categorizes existing research by consensus mechanisms, data protection, and blockchain types. The review confirms blockchain's benefits, like transparency and tamper resistance, but also highlights persistent challenges in scalability, privacy, and interoperability. [2]

Minango et al. (2024) focused on the practical application of blockchain for academic certificates. They detailed a smart contract-based system using decentralized nodes for credential issuance and verification. Validated with real institutional data, their approach demonstrated enhanced accessibility, data integrity, and transparency,

showing that blockchain can streamline administration while protecting user privacy. [3]

A. K. C et al. (2024) introduced a Blockchain-Based Certificate Verification and Issuer Validation System to detect fake physical certificates. Their novel hybrid framework combines the immutability of blockchain with instantaneous validation from issuing authorities. Experimental results showed this system is highly effective in identifying fake credentials and preventing fraud. [4]

Stefan-Robert and Butincu (2024) proposed a decentralized, blockchain-based platform for managing and distributing academic credentials. Their design leverages smart contracts to ensure decentralized authority and data transparency. The architecture notably prioritizes user confidence, scalability, and compliance with European data protection regulations. [5]

Collectively, these studies show blockchain significantly impacts academic credential management. While existing frameworks promise enhanced security and efficiency, significant challenges remain, including data privacy, scalability, and inter-institutional consistency. To achieve broad adoption, future research must prioritize developing blockchain systems that are interoperable, private, and compliant with regulations.

Table 1: Survey of Blockchain Papers

Research Article	Objective	Methodology	Outcomes	Limitations
Development of Blockchain-Based Academic Credential Verification System.[1]	Design a secure blockchain verification system for credentials.	Developed a permissioned blockchain (Hyperledger Fabric) with a web interface.	Achieved fast, secure, and tamper-proof verification.	Interoperability limits, partial data, and privacy constraints.
A Systematic Literature Review on Blockchain-Based Systems for Academic Certificate Verification. [2]	Analyze blockchain verification frameworks; identify models and challenges.	Systematic review of 80+ studies, evaluating architectures and consensus methods.	Defined a taxonomy of blockchain solutions for authenticity and fraud prevention.	Most models are prototypes; limited real-world adoption; regulatory barriers.

Exploring the Use of Blockchain for Academic Certificates: Development, Testing, and Deployment. [3]	Design and test a blockchain framework for issuing/verifying academic certificates.	Built a prototype using Ethereum smart contracts and IPFS; tested for performance/security.	Enabled tamper-proof, transparent, instant verification; improved trust.	Scalability issues, high gas fees, and limited user adoption due to complexity.
Detection of Fake Physical Certificates using a Blockchain-based Certificate Verification and Issuer Validation System. [4]	Develop a blockchain system to verify physical certificates and validate issuers.	Used a blockchain ledger; QR codes/hash mapping to link physical/digital records.	Enhanced trust and transparency; reduced fake certificates; provided tamper-proof records.	High initial deployment cost; scalability issues; requires widespread issuer adoption.
Decentralized Blockchain-based Platform for Managing and Issuing Academic Certificates. [5]	Design a decentralized platform for secure certificate management without a central authority.	Used blockchain for issuance/storage with smart contract-based access control.	Decentralized management increased transparency and security.	Complex integration with existing systems; relies on stakeholder technical literacy.

AI-powered solutions are increasingly used to reduce human bias, optimize job matching accuracy, and accelerate resume evaluation. Recent research utilizes deep learning, natural language processing, and data-driven models to improve the overall effectiveness and fairness of hiring procedures.

Rezaeipourfarsangi and Milios (2023) introduced an AI-powered resume-job matching framework using deep neural networks. Their ranking-based model captures semantic similarity between resumes and job descriptions, significantly outperforming traditional keyword-matching methods in accuracy and relevance for automated candidate selection. [6]

Abhishek et al. (2025) developed an intelligent resume screening tool that combines machine learning and NLP to evaluate resumes, predict job fit, and suggest candidates. The study highlights the tool's ability to improve impartiality and matching accuracy while simultaneously reducing recruiter effort. [7]

Dadaboyev et al. (2025) conducted a systematic review on AI in recruitment, focusing on its impact

on the effectiveness and dependability of hiring. They also identified critical problems for future research, including algorithmic bias, transparency concerns, and moral conundrums. [8]

Kong et al. (2021) provided an overview of AI-driven recruitment tools, analyzing their features, usability, and ethical considerations. They emphasized that transparent, human-monitored AI systems are crucial for upholding accountability and trust in hiring practices. [9]

Taken as a whole, these studies highlight the evolution of AI in hiring, from basic document-ranking to intelligent, recommendation-driven systems. They demonstrate that AI enhances productivity and data-driven decision-making in candidate selection. However, persistent issues such as bias, data privacy, interpretability, and human-AI cooperation remain crucial topics for future research.

Table 2: Survey of AI Papers

Research Article	Objective	Methodology	Outcomes	Limitations
AI-powered Resume-Job matching: A document ranking approach using deep neural networks. [6]	To improve resume-job matching using deep learning ranking models.	DNN-based document ranking model for resumes and job descriptions.	Higher matching accuracy than traditional methods; scalable.	Requires high-quality labeled data; limited interpretability.
Developing an Intelligent Resume Screening Tool With AI-Driven Analysis and Recommendation Features. [7]	Build an AI-driven resume screening and recommendation tools.	Used NLP for parsing; ML algorithms for scoring and ranking.	Increased screening efficiency and accuracy; reduced HR workload.	Risk of algorithmic bias; struggles with creative/unstructured resumes.
Role of artificial intelligence in employee recruitment: systematic review and future research directions. [8]	Review literature on AI in recruitment and find research gaps.	Systematic literature review and thematic analysis.	Identified AI recruitment trends (e.g., chatbots, predictive analytics).	Limited to published studies; may miss industry-only applications.
AI-Assisted Recruiting Technologies: Tools, Challenges, and Opportunities. [9]	Discuss AI recruitment tools, challenges, and opportunities.	Exploratory study summarizing tools; identified ethical/communication issues.	Overviewed emerging tools and their impact; highlighted adoption challenges.	Descriptive, not experimental; lacks empirical data. Findings may date quickly.

3. PROPOSED SYSTEM

3.1 System Overview and Architecture

SPCP is implemented as a highly scalable microservice-based Software as a Service (SaaS) solution. The architecture is designed to handle multi-tenancy colleges and companies and provide high resilience owing to the modular separation of concerns, as shown in the block diagram below.

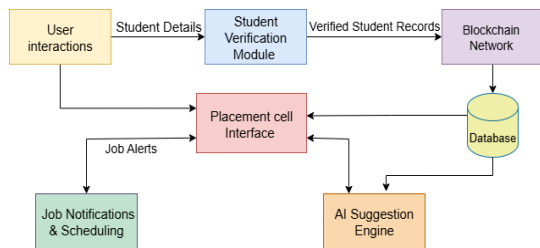


Figure 1 Architectural Style: Microservices

1. **Client Tier:** Contains user-visible interfaces, like the Next.js web app and admin dashboards. This tier handles only presentation and user input.

2. **Application Tier:** This is the functional core, built on Node.js and accessed via an API gateway. Each service handles specific business logic, allowing independent deployment and scaling:
 - a. **Profile Service:** Manages user registration and digital profiles.
 - b. **Job Service:** Handles recruiter job postings and vacancy queries.
 - c. **Matching Service:** Runs AI-driven shortlisting and job recommendations.
 - d. **Credential Service:** Issues and verifies blockchain-based credentials.
 - e. **Admin Service:** Provides core administrative and analytical functions.
3. **Data and External Services Tier:** This tier includes all persistence layers and specialized third-party integrations:
 - a. **Database (DB):** Stores transactional data, profiles, and application data.
 - b. **AI/ML Module:** Executes complex machine learning pipelines for resume parsing and match scoring.

- c. **Blockchain Module:** Serves as the immutable ledger for credential hashes, ensuring data integrity.

This type of decoupled architecture ensures that modules such as the AI/ML Module and the Blockchain Module are independently updatable or scalable without hindering the core placement functionalities.

3.2 Functional Modules and Data Flow

This section details the SPCP's main functional components, their operations, and data flow..

3.2.1 Identity and Profile Management

This module handles user onboarding and secure profile maintenance. Students register via the Web App, with the **Profile Service** storing their verified information (including KYC status) in the **Database**.

Dataflow: Student Data → Profile Service → Database.

3.2.2 Job and Application Engine

This module manages the job posting and application lifecycle. Recruiters post jobs via the Web App, which the **Job Service** stores in the **DB**. When students apply, the Job Service records the application and triggers the matching process.

Data Flow: Recruiter Job Data → Job Service → DB.

Student Application → Job Service → Matching Service.

3.2.3 AI Matching and Recommendation Service

This module uses predictive analytics for efficient matching. When an application is submitted, the **Matching Service** sends job and profile data from the **DB** to the **AI/ML Module**. This module runs algorithms (e.g., NLP resume parsing) to generate a match score, which is then stored back in the DB.

Data Flow: Job/Profile Data (from DB) → Matching Service → AI/ML Module → Match Score → DB.

3.2.4 Credential Verification Service

This security layer uses blockchain to ensure data integrity. After an administrator verifies a document, the **Credential Service** computes its cryptographic hash and anchors it to the **Blockchain Module**. Recruiters can then verify credentials by matching a document's hash against the immutable one stored on the blockchain

Data Flow: Document Hash → Credential Service → Blockchain Module.

3.2.5 Admin and Analytics

This module provides management, monitoring, and interoperability. The **Admin Service** fetches data from the **DB** to populate Admin Dashboards with KPIs and audit logs.

3.3 Text Mining with Neural Network

Text mining, closely linked with NLP techniques like text classification and similarity, can be applied to study person-job fit using textual data. Due to advancements in performance, researchers increasingly apply deep learning to text mining. Unlike traditional methods requiring heavy feature engineering, deep learning models learn effectively end-to-end from large-scale text. CNNs and RNNs are two of the most popular deep learning architectures successfully used for various NLP problems. [10]

4. EXPERIMENTAL SETUP

Since the blockchain component is currently under architectural development, our practical experiments focused on validating the "intelligence" pillar of our platform. We aimed to prove that our AI agent could accurately ingest raw student data and structure it for the matching engine without manual intervention.

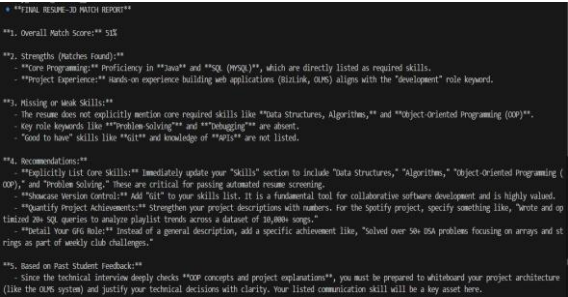
4.1 Software and Tools

We developed the prototype in a Python environment, choosing LangChain as our primary orchestration framework to manage the interaction between our data and the model. To handle the control flow ensuring the agent maintains state and follows a

logical reasoning path we utilized LangGraph. For the underlying processing power, we integrated Google’s Gemini API. We selected Gemini specifically for its robust natural language understanding and large context window, which is critical when processing detailed academic CVs that often exceed standard token limits.

4.2 Dataset and Methodology

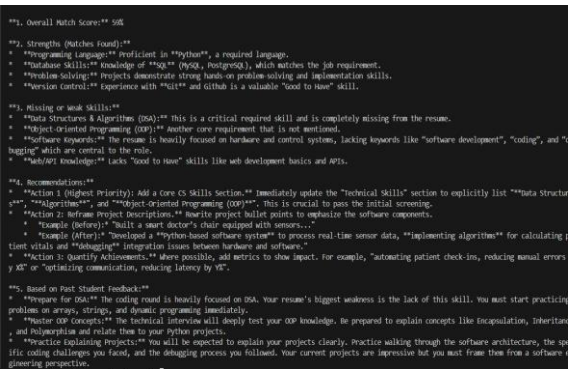
To stress-test the system, we compiled a dataset of student resumes featuring a wide variety of layouts, terminologies, and file structures. The challenge was to take this "unstructured chaos" where one student might list "Skills" and another "Technical



Proficiencies" and standardize it. The agent was tasked with reading these documents and mapping the information to a strict JSON schema that our database can index for job matching.

4.3 Output and Results

The experimental runs confirmed the viability of our approach. The LangGraph-based agent successfully navigated the complex document structures, accurately distinguishing between educational history, project work, and technical skills. As shown in the figures below, the system effectively converted



raw text into a structured JSON output. This confirms that the critical first step of the recruitment pipeline data ingestion and profiling can be fully automated with high accuracy.

Figure 1: Execution logs of the LangGraph AI Agent interacting with the Gemini API

Figure 2: The final structured JSON output generated by the model from a raw resume.

5. FUTURE WORK & DIRECTIONS

5.1 Advanced AI and Machine Learning Enhancements

Explainable AI (XAI) Integration: To address limited AI transparency, future releases will incorporate a comprehensive XAI framework. Using interactive dashboards, this framework will explain algorithmic decisions—such as job ratings, recommendations, and placement predictions—in an understandable way. This increased transparency is intended to build user trust and enable better decision-making. [11] [12]

Multi-modal AI processing: Video interviews, portfolio presentations, and project demonstrations will all be processed using advanced AI capabilities. During virtual interviews, computer vision algorithms interpret nonverbal cues to give candidates a more thorough evaluation. [13]

5.2 Blockchain Technology Evolution

To ensure scalability and interoperability, universities should introduce blockchain systems gradually, starting with simpler services before progressing to degree verification. This process requires evaluating data transfer from current databases, testing major blockchain cloud platforms, and securing data conversion. Achieving optimal performance and scalability depends on integrating both on-chain and off-chain storage. Furthermore, cross-chain technology is crucial for enhancing blockchain interoperability, as it facilitates transactions between chains without third-party interfaces, thereby improving efficiency and reducing fragmentation. [14] [17]

Using blockchain for diploma verification could significantly reduce administrative work related to security, privacy, and verification, while also decreasing forgeries by enabling simple

confirmation. However, this implementation faces challenges. Creating and rescinding diplomas is difficult because blockchain security relies on unique hash values; a revocation alters the hash, which can cast doubt on the student's identity. More research is essential on securely integrating blockchain databases with current systems. Any practical solution must also address data privacy laws like GDPR, as blockchain's inherent immutability may conflict with regulations such as the "right to be forgotten." [15] [17]

To guarantee blockchain's resilience, more empirical research is needed from the standpoints of sustainability and trust. While trust is fostered by the technology's transparent, safe, and unchangeable services, a key challenge remains: how to assess or confirm the legitimacy of an organization issuing blockchain credentials. The literature review also indicates that blockchain provides extra advantages, such as enabling cutting-edge online teaching strategies that incorporate group projects. [16] [17]

This puts the SPCP in a good position to develop into a cutting-edge platform that will not only adjust to new technologies but also assist in addressing some of the core issues with contemporary campus placement and recruitment procedures.

6. CONCLUSION

By methodically filling research gaps, the SPCP model guides the enhancement of campus recruitment systems. Although initial implementation involves advanced technology and a steep learning curve, the long-term benefits—improved placement rates, reduced credential fraud, and greater stakeholder trust—will greatly exceed these costs. The SPCP framework is well-positioned to set a new standard for safe, effective, and fair campus hiring, laying the groundwork for the next education-to-employment transformation.

The SPCP radically reimagines placement management by offering real-time analytics, intelligent automation, verifiable credential authenticity, and robust privacy protections in traditionally flawed areas. This integrated platform empowers students, recruiters, and institutions through data-driven decisions, process clarity, and reduced administrative burdens. It is technologically advanced, integrating blockchain and AI/ML to meet institutional needs, and progressive, readying it for

innovations like decentralized identity and federated learning.

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